

Gradient Descent Basics

Gradient descent minimizes a loss function by moving opposite the gradient.

Learning rate controls step size and convergence behavior.

Optimization Hyperparameters Table

Term	Description
Learning Rate	Controls step size in each optimization update
Momentum	Uses velocity to smooth updates over iterations

Loss Curve Illustration

The image represents decreasing loss over training epochs.



Optimization Notes

Batch gradient descent computes gradients over the full dataset.

These notes are relevant to optimization-focused study prompts.

Optimization Notes

Stochastic gradient descent updates per sample and can escape shallow minima.

These notes are relevant to optimization-focused study prompts.

Optimization Notes

Mini-batch gradient descent balances stable gradients and training speed.

These notes are relevant to optimization-focused study prompts.